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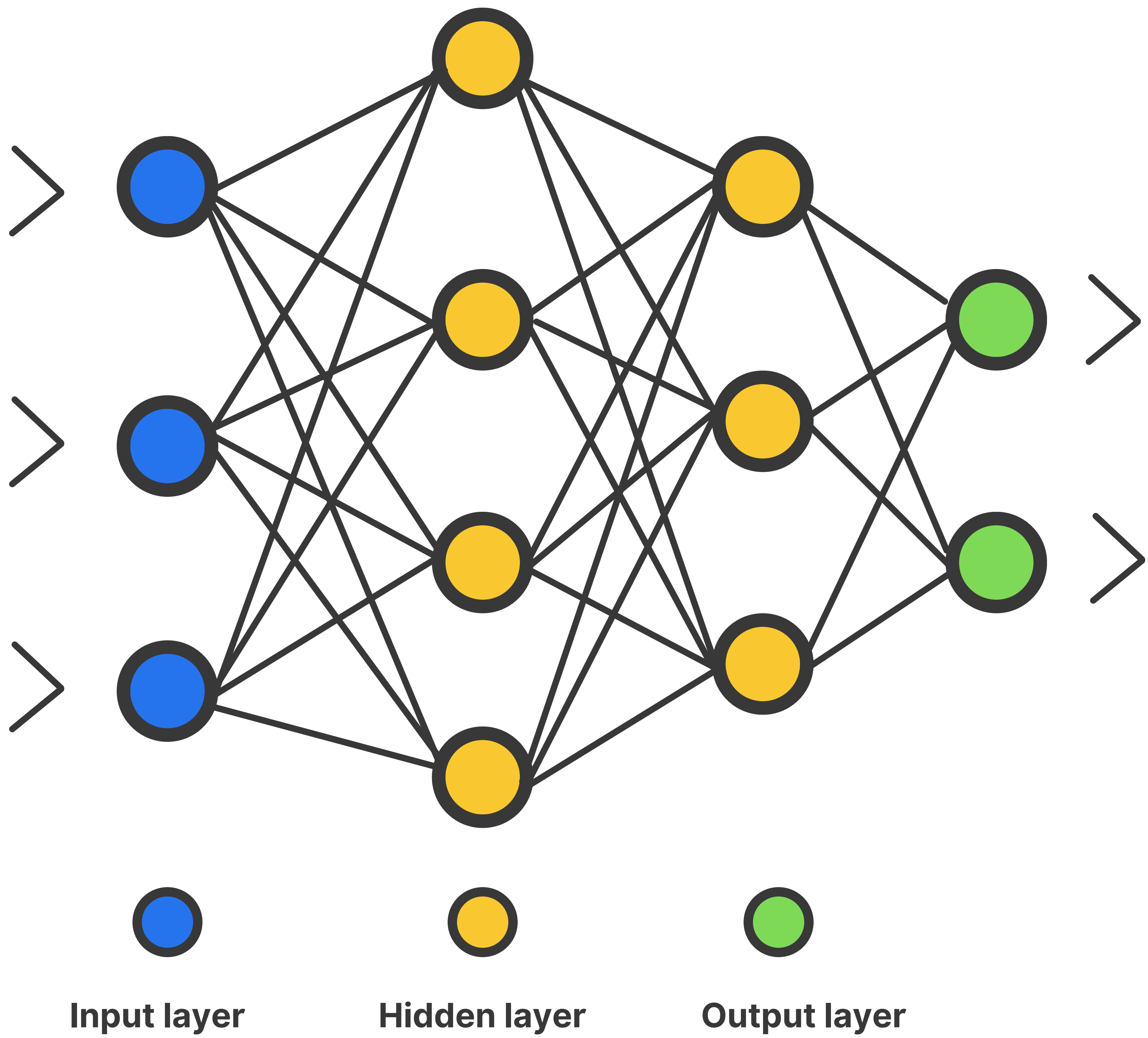
DAY 4 OF DAILY DEEP-DIVE SERIES

A Guide to Neural Networks



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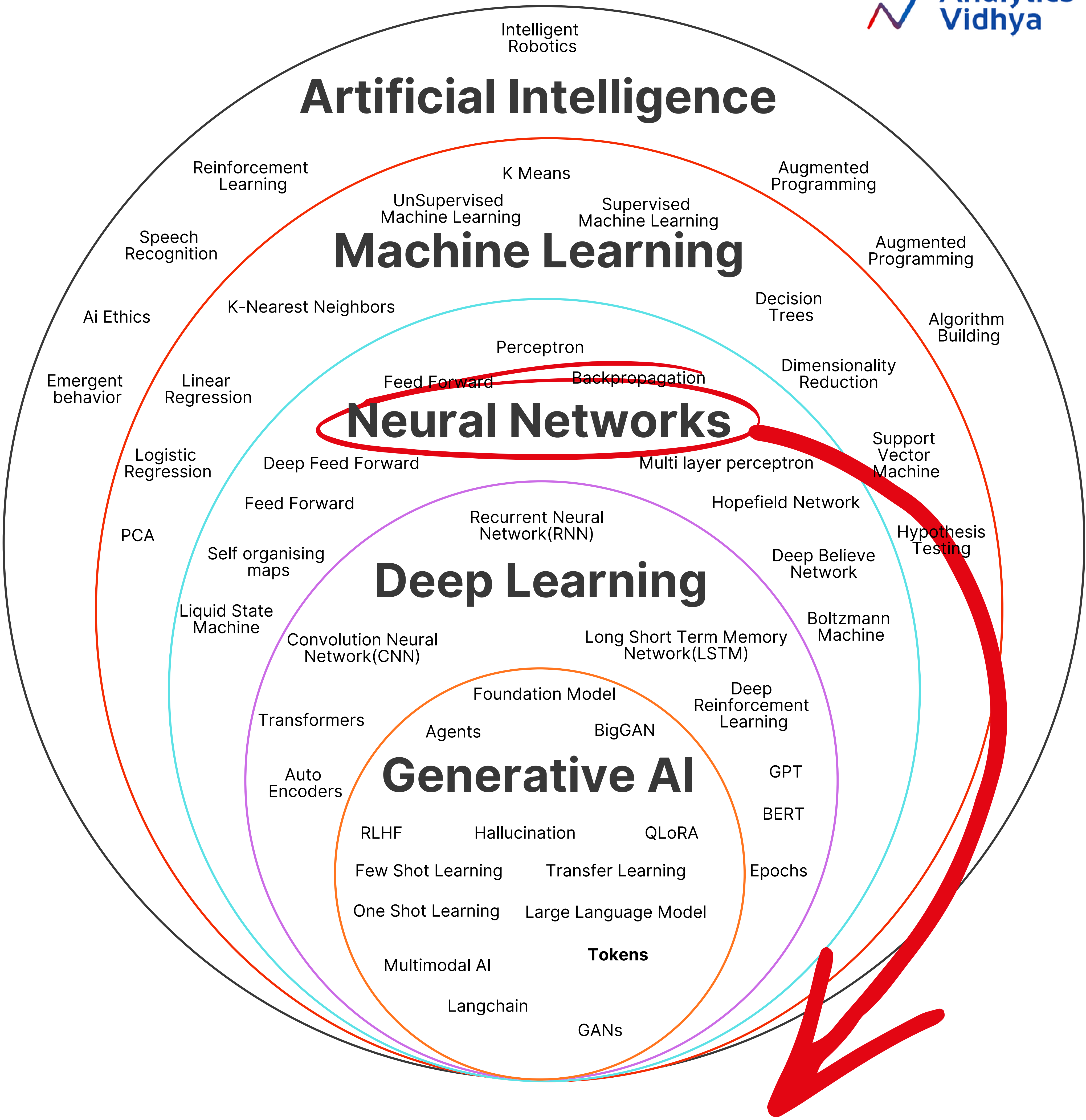

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What is a Neural Network?

A neural network is a computational model inspired by the **human brain's network of neurons**.

It consists of an interconnected group of artificial **neurons** that process information using a connectionist approach to computation.

Basic Components

- **Neurons**: The basic units of computation that mimic biological neurons.
- **Layers**: Groups of neurons organized into layers—Input, Hidden, and Output layers.

Feedforward and **backpropagation** are two fundamental phases in the training and operation of neural networks. Understanding these two concepts is crucial for understanding how neural networks learn from data.

Feedforward

The initial phase where the data flows from the input layer through hidden layers to produce the output.

Backpropagation

The learning phase where the model adjusts its weights and biases based on the error of its predictions.

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Learning Algorithm

1. Initialize random weights for all edges (connections).
2. For each input:
   a. Perform a feedforward pass.
   b. Compute the loss between the predicted and true values.
   c. Perform backpropagation to update the weights.
3. Repeat step 2 for multiple epochs or until other stopping conditions are met.

Python Example with TensorFlow

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import mnist

# Load dataset
(train_images, train_labels), (test_images, test_labels) = mnist.load_data()

# Preprocess the data
train_images = train_images.reshape((60000, 28 * 28)).astype('float32') / 255
test_images = test_images.reshape((10000, 28 * 28)).astype('float32') / 255

# Convert labels to categorical
train_labels = tf.keras.utils.to_categorical(train_labels)
test_labels = tf.keras.utils.to_categorical(test_labels)

# Define the model
model = models.Sequential()
model.add(layers.Dense(512, activation='relu', input_shape=(28 * 28,)))
model.add(layers.Dense(10, activation='softmax'))

# Compile the model
model.compile(optimizer='rmsprop',
              loss='categorical_crossentropy',
              metrics=['accuracy'])
```

Python Example with TensorFlow

```python
# Train the model
model.fit(train_images, train_labels, epochs=5, batch_size=128)

# Evaluate the model
test_loss, test_acc = model.evaluate(test_images, test_labels)
print(f'Test accuracy: {test_acc}')
```

By understanding these basic elements and practicing through coding, you'll develop a better grasp of neural networks and how they can be applied to solve complex problems.